Low-Rank Adapters for Small Sentiment Classifiers: A Synthetic Demo

Aster Research Demo

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Abstract

This manuscript belongs to the Aster harness demo. Every run in it was produced by a seeded simulator, not by training a model, so the numbers exercise the workflow rather than support a scientific claim.

1 Question

How much validation accuracy does a LoRA adapter [1] gain from a higher rank when a small encoder [3] is fine-tuned on a five-class sentiment task [4], and which learning rate keeps training stable?

2 Method

All runs use AdamW [2], batch size 32 and 600 optimizer steps. Each run was launched with the research harness, which records the code at launch, the exact command and every metric event.

3 Results

Table 1 reports validation accuracy after 600 steps. With rank 16 the adapter reaches 82.9% validation accuracy. Rank 64 raises it to 86.4% on the same split. Moving from rank 16 to rank 64 therefore adds about 2.9 points.

Rank	Trainable parameters	Val. accuracy
4	0.3M	80.0%
16	1.2M	82.9%
64	4.7M	86.4%

Table 1: LoRA rank ablation (learning rate 3×10^{-4}, seed 2).

In the learning-rate sweep, 3×10^{-4} reaches 83.2% validation accuracy, while 10^{-3} diverges at step 300. At 10^{-4} the validation loss is still 1.047 after 600 steps.

4 Limitations

A single seed per configuration cannot separate the rank effect from run-to-run noise.

References

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[3] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

[4] Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of EMNLP*, 2013.